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Summary

Dancing with the Stars (DWTS), as a leading international television franchise, has long attracted global audiences through its integration of professional artistry and public participation. In this study, we build an evaluative model to reconstruct latent fan voting distributions, a simulation framework to assess the impact of scoring regimes, and a strategic system to optimize competition balance.

Firstly, after data preprocessing and feature engineering, we establish the **Bayesian Temporal Estimation (BTE) Model** to reconstruct fan votes. This model constructs concentration parameter vector $\boldsymbol{\alpha}_w$ which measures contestant latent competitiveness as prior distribution and utilizes the **Metropolis-Hastings (MH) sampling algorithm** to derive posterior estimates. To validate the reliability of latent variables, we introduce and calculate the **Elimination Match Rate (EMR)=0.61, Weekly System Certainty (WSC)=0.732,** indicating that our BTE model provides a high-fidelity.

Secondly, leveraging the posterior distributions, we construct the **Contest Rule Simulation Model with Uncertainty Propagation (CRSM-UP)** to evaluate the systemic resilience of different elimination regimes. Through **Counterfactual Inference**, we execute a multi-dimensional comparative analysis: a global assessment via the **Fan Influence Index ($\mathcal{F}$)** yields a significant $t-statistic\ of\ 11.939\ (p < 0.001)$; for controversial contestants, a simulation using the **Bradley-Terry Probabilistic Model** reveals a survival rate variance exceeding **10%**; and a evaluation of **Technical Defense Strength (TDS)** identifies a **30% gap** between the two scoring operators.

Collectively, these findings demonstrate that while the Rank-based system prioritizes professional performance, the Percentage-based regime significantly amplifies fan influence**.** Consequently, we recommend an optimized mechanism combining the **Rank-based System with a "Save One" principle** to harmonize competitive integrity with audience engagement.

Thirdly, to decode the divergent evaluation criteria between judges and fans, we establish the **SHAP-enhanced Gradient Boosting Decision Tree (SE-GBDT) Model.** We employ **XGBoost** as the predictive engine and integrate **SHAP** (Shapley Additive Explanations) values to quantify marginal contribution of each feature. Significantly, fan voting is mainly driven by narrative **momentum**, and judge voting is driven by **technical consistency**, but they hold consensus on survival of the fittest and age. Additionally, we find more effects in 'All-Star' season 15 such as the Mattew effect and the Champion's halo.

Finally, we created a **Dynamic Growth-Circuit Breaker (DGCB) System** which combines **dynamic composite score** adding **Growth Bonus** with the **'Circuit Breaker'**, targeting the triangle of 'fairness, excitement and profitability.'

In addition, we perform sensitive analysis, discuss the strengths, weaknesses and future of our model and report our findings in a memo to the DWTS's production team.

Keywords: DWTS; Bayesian Temporal Estimation; Counterfactual Inference; XGBoost; SHAP; Dynamic Growth-Circuit Breaker system

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS), the U.S. version of Britain's "Strictly Come Dancing" franchise, has grown into a global cultural phenomenon with localized adaptations in over 40 countries. Spanning 34 seasons, this flagship reality show blends celebrity appeal with professional ballroom dancing, aligning with the global trend of interactive entertainment that emphasizes audience connection with public figures.

The core of the show's enduring popularity lies in its dual-evaluation system: expert judge scores (assessing technical proficiency with inherent subjectivity) and fan votes (driven by celebrity popularity and emotional resonance). Hailed as a "paradigm of balanced reality competition," this design balances artistic rigor with audience engagement. To address recurring controversies over "technical underperformers advancing via fan support," the program has undergone three major rule revisions: rank-based voting, percentage-based voting, and a return to rank-based voting, with supporting elimination adjustments.

However, there exists intense academic and public debate surrounding this dual system. Proponents, such as media scholar Hill, argue that the combination of judge scores and fan votes "reconciles institutional authority with participatory culture," emphasizing that the system's inherent flexibility to adapt to fairness concerns contributes to sustained viewership appeal [1]. Critics, by contrast, contend that the system's "fairness" is a constructed media illusion. For instance, Couldry's research on reality TV transparency points out that the non-disclosure of raw fan vote data— an industry practice to preserve suspense— "blocks empirical verification of whether votes align with performance," rendering claims of "balanced evaluation" untestable [2].

Amid growing demands for media accountability, the show offers a pivotal case to explore the interplay between professional evaluation, public participation, and televised competition rules. Investigating its voting mechanism resolves the show's own controversies and provides actionable insights for dual-evaluation system design in global reality TV.

1.2 Restatement of the Problem

Based on the limited observational data, we abstract the *Dancing with the Stars* evaluation problem into four progressive mathematical modeling problems:

Problem 1: Reconstruction of Latent Fan Votes ： Solve the inverse problem to estimate the hidden distribution of fan votes from elimination constraints, quantifying the consistency and certainty of these estimates to establish a data benchmark.

Problem 2: Mechanism Simulation & Assessment ： Conduct counterfactual simulations using the estimated data to compare the sensitivity of Rank-Based vs. Percentage-Based systems and evaluate the efficacy of the "Judge Save" rule in handling controversial outcomes.

Problem 3: Driver Attribution Analysis ：Construct predictive models to quantify the nonlinear impacts of features (e.g., pro partners, contestant background) on judges versus fans, identifying the structural divergence in their preferences.

Problem 4: System Optimization Design ： Design a novel evaluation system (DGCB)

that balances fairness (professional merit) and engagement (fan participation), aiming to systematically mitigate technical deficit anomalies while retaining commercial viability.

1.3 Our Work

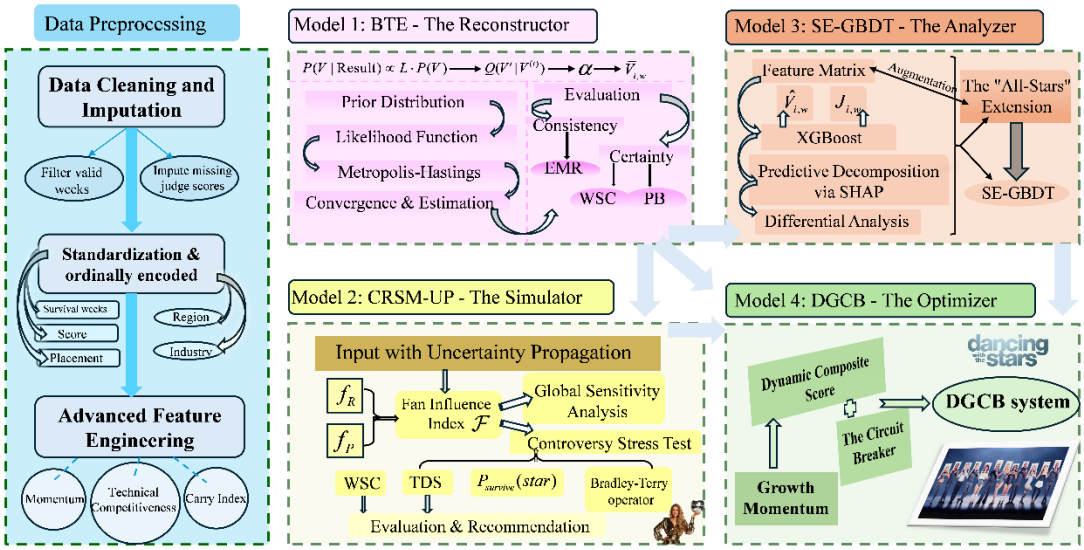


Figure 1: Our Work

2 Assumptions and Justifications

Assumption 1: We assume that fan preferences exhibit temporal inertia, implying that fan voting behavior exhibits memory. Specifically, the vote distribution received by a contestant in $Week_t$ is strongly correlated with their competition performance and underlying popularity base in $Week_{t-1}$, with no abrupt, unsubstantiated fluctuations. This assumption establishes the temporal dependence required for the MCMC sampling procedure in our BTE model.

Assumption 2: We assume that judges hold probabilistic preferences governed by technical merit. In bottom_two save scenarios, judicial decisions are not implemented as rigid deterministic rules, but display stochasticity that correlates with the differential between contestants' technical scores. Concretely, the probability that a judge saves a given contestant increases monotonically with their technical score advantage, and this relationship is characterized by the Bradley-Terry probabilistic model.

Assumption 3: We assume that the final voting share admits an additive decomposition into the marginal contributions of individual features within our SE-GBDT model. Although we adopt non-linear tree-based models to capture feature interactions, these explanatory features are sufficient to account for the majority of variance in voting outcomes.

Assumption 4: With our DGCB Model, it is assumed that when the technical circuit breaker is triggered, the system can operate independently. Our DGCB system is fully insulated from external interference such as fan voting, and the entire adjudication process is

governed exclusively by a predefined mathematical threshold, free from any producer manipulation.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
i	Index of contestants	-
w	Competition week	Week
s	Season of the competition	Season
$V_{i,w,s}$	Proportion of fan votes for contestant i in season s and week w	
$\hat{V}_{i,w}$	Optimal estimate of fan vote proportion for contestant i in week w (mean of effective MCMC samples)	
$J_{i,w}$	Average judge score of contestant i in week w	
S_i	Comprehensive competitiveness score of contestant i	
$R(X)$	Descending rank of indicator X (1 = highest)	Rank (1,2,...,n)

4 Data Description and Preprocessing

4.1 Data cleaning and imputation

Handling N/A and Zero Values: First, we defined a valid competition window for each contestant by extracting their elimination status from the results variable. Subsequently, all zero-score records occurring post-elimination were removed. Finally, the dataset was refined by purging null entries (N/A) associated with non-broadcast periods, thereby aligning the temporal dimensions across varying seasonal structures.

4.2 Standardization and ordinally encoded

Table 2: Standardization and Ordinal Encoding of Core Variables

Variable Name	Standardized Formula/ Brief Description of Encoding Method
result （Survival Weeks）	$Z_{survive} = \frac{survival_weeks-\mu}{\sigma}$
Placement – NPI	$Z_{NPI} = \frac{NPI-\mu}{\sigma}$
weekX_judgeY_score	$Z_{score} = \frac{raw_score-\mu}{\sigma}$
industry	Ordinal encoding: (Exposure score + Dance relevance score)/2 Exposure: High=3, Medium=2, Low=1 Dance relevance: High=3, Medium=2, Low=1
region	Ordinal encoding: American: Top10 states=3, 11-30th states=2, Others=1 Non-American: Top10+English-speaking=3, Top10 non-English/11-50th English=2, Others=1

4.3 Feature Engineering

Feature engineering addresses raw data limitations (fragmented information, uninterpretable text, incomplete data) by converting implicit rules (e.g., growth trends, popularity accumulation) into explicit indicators, supporting subsequent modeling and improving model interpretability and prediction accuracy.

Table 3: Derived Features for Feature Engineering

Variable Name	Formula	Description
$SP_{i,w}$ (Explicit Popularity)	$SP_{i,w} = \sum_{k=1}^{w-1\sum_{i,k}\text{Result}_{i,k}} max$	Accumulate the explicit popularity base from the contestant's past performance
CI_i (Carry Index)	$CI_i = \sum \frac{NPI_i - E[NPI \mid Initial_Score_i]}{Tenure}$	Quantify the partner's "carrying capacity"
$RM_{i,w}$ (Relative Momentum)	$RM_{i,w} = (J_{i,w} - J_{i,w-1}) - \frac{1}{n}\sum_{j=1}^{n}(J_{j,w} - J_{j,w-1})$	Reflect the contestant's technical progress rate advantage over competitors in the same period
$TC_{i,w}$（Technical Competitiveness）	$TC_{i,w} = \frac{\sum_{k=1}^{w} standardized_score_{i,k}}{\sum_{k=1}^{w} \text{Mean}(standardized_score_{i,k})}$	Cumulative relative technical strength up to the current week

5 Problem 1: Reconstruction of the Invisible

5.1 Model Selection Basis: A Bayesian Inverse Approach

Since fan voting data is unobservable, we treat it as a latent variable estimation problem. We propose the **Bayesian Temporal Estimation (BTE)** model and deliver dedicated metrics for measuring consistency and certainty associated with fan voting outcomes. Unlike deterministic fitting, this framework allows us to:

Integrate Heterogeneous Information: Fuse hard constraints (elimination results) with soft priors (historical popularity/performance features).

Quantify Uncertainty: Instead of a single value, we obtain a probability distribution for each contestant's vote share, enabling the risk assessment required in Problem 2.

5.2 Model Construction Process

5.2.1 Construction of Informative Priors

Leveraging the feature engineering from Section 4 (Carry Index, Momentum, etc.), we construct an informative prior to anchor the inference. Let $\boldsymbol{V}_w = \{V_{1,w}, \dots, V_{n,w}\}$ denote the vector of fan vote proportions for n contestants in week w, subject to the simplex constraint $\sum V_{i,w} = 1$.

We assume $\boldsymbol{V}_w$ follows a **Dirichlet Distribution**, parameterized by a hyperparameter vector $\boldsymbol{\alpha}_w$:

$$P(\boldsymbol{V}_w | \boldsymbol{X}_w) = \text{Dir}(\mathbf{V}_w | \kappa \cdot \boldsymbol{\alpha}_w) \tag{1}$$

Where the concentration parameter vector $\boldsymbol{\alpha}_w$ is a linear combination of the normalized feature matrix$\boldsymbol{X}_w$:

$$\alpha_{i,w} = \boldsymbol{w}^T \cdot [CI_i, RM_{i,w}, TC_{i,w}, SP_{i,w}]^T \tag{2}$$

The calculated weights are $\omega_1 = 0.04\ (CI), \omega_2 = 0.03\ (RM), \omega_3 = 0.29\ (TC), \omega_4 = 0.64\ (SP)$, satisfying $\omega_j = CV_j / \sum CV$.

Here, $\kappa \sim \text{Gamma}(2,0.1)$ is a hierarchical hyperparameter controlling the stiffness of the prior (i.e., how much the model relies on historical features vs. current elimination data).

5.2.2 Adaptive Likelihood Function

The likelihood function $L(D|\boldsymbol{V}_w)$ bridges the latent votes to the observed elimination results$D$. Crucially, our model adapts to the show's changing rules by defining two distinct scoring kernels:

Comprehensive Score Definition (S_i):

$$S_i(\boldsymbol{V}) = \begin{cases} -(R(J)_i + R(V)_i) & \text{if Season} \in \text{Rank-Based} \\ \frac{J_i}{\sum J} + V_i & \text{if Season} \in \text{Percentage-Based} \end{cases} \tag{3}$$

The Elimination Probability (Plackett-Luce Formulation):

We model the elimination event as a probabilistic process where contestants with lower composite scoresS_i face higher risks. The likelihood of the observed set of eliminated contestants E is:

$$L(E|\boldsymbol{V}_w) \propto \prod_{k \in E}^{\prod} \frac{exp(-\lambda S_k(\boldsymbol{V}_w))}{\sum_{j \in \mathcal{R}}^{\sum j_w} exp} \tag{4}$$

Where $\mathscr{R}$ is the set of contestants at risk, and λ (optimized λ=1.2 via cross-validation)is the temperature parameter controlling the determinism of the elimination rule.

5.2.3 Posterior Inference via MCMC

According to Bayes' Theorem, the posterior distribution is:

$$P(\boldsymbol{V}_w|E, \boldsymbol{X}_w) \propto L(E|\boldsymbol{V}_w) \cdot P(\boldsymbol{V}_w|\boldsymbol{X}_w) \tag{5}$$

Since this analytical form is intractable, we employ the **Metropolis-Hastings (MH)** algorithm to sample from the posterior. The detailed sampling procedure is outlined in **Algorithm.**

Algorithm: Bayesian Temporal Estimation (BTE) Model for Fan Voting

Input: $CI_i, RM_{i,w}, TC_{i,w}, SP_{i,w}$ ***Placement, Active weeks and schedule_type***

Output: $\widehat{V_{i,w}}$ ***and*** $\sigma^2_{i,w}$

```
# A. Initialize prior parameters
α_{i,w} ← 0.1·CI_i + 0.05·RM_{i,w} + 0.7·TC_{i,w} + 0.15·SP_{i,w}
μ_{w,i} ← Normalize(α_{1:n,w})
κ ~ Gamma(2,0.1)
P(V_w) ~ Dirichlet(κ·μ_w)
# B.Construct softened likelihood function
if schedule==ranking:
        S_i ← −(R(J)_i + R(V)_i)
if schedule==percentage:
        S_i ← StandardizedScore(i)
for each episode with eliminated set E:
        L(V|E) ← ∏_{k∈E} exp(−λS_k) / Σ_{j∈E\{k}} exp(−λS_j), λ = 1.2
if judges' decision exists (season>=28):
        L(V|E) ← L(V|E) · exp(w · J_{k_1})
# C. Perform Metropolis-Hastings sampling
Initialize V^(0) ← μ_1
for t ← 0 to T−1 do:
        ## Logit-transform and propose new point
        y^(t) ← Logit(V^(t))
        y' ← y^(t) + ε, ε ~ N(0, σ_0^2 I)
        V' ← Softmax(y')
        ## Compute acceptance probability
        α ← min(1, L(V'|Result)P(V') / L(V^(t)|Result)P(V^(t)))
        if u ~ U(0,1) < α:
                V^(t+1) ← V'
        else:
                V^(t+1) ← V^(t)
        ## Check convergence every 1000 iterations
        if t mod 1000==0:
                compute R̂ across 4 chains
                if max(R̂) < 1.05:
                        break
# D. Output estimation
Let {V^(B), ..., V^(T)} be post-burn-in samples:
        V̂_{i,w} ← 1/(T−B) Σ_{t=B}^{T} V_i^(t)
        V̂_{i,w} ← 1/(T−B) Σ_{t=B}^{T} V_i^(t)
return V̂_{i,w}, σ^2_{i,w}
end
```

5.3 Convergence Analysis and Estimation Results

5.3.1 MCMC Convergence Diagnostics

To ensure the reliability of our estimates, we ran 4 independent Markov chains with 20,000 iterations each. Figure visualizes the trace plot for a sample week (Season 5, Week 10).

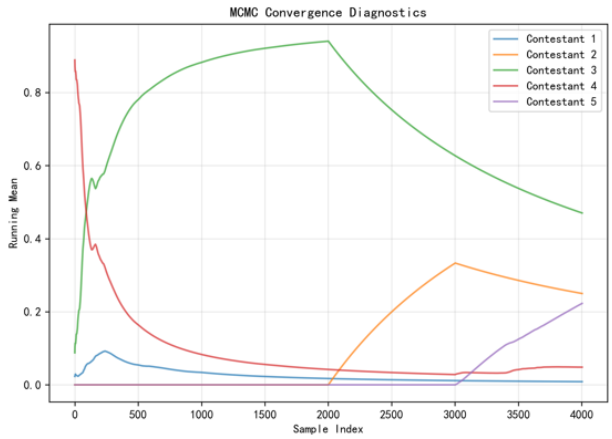


Figure 2: MH Sampling Convergence and Vote Estimation (Season 5 Week 10)*

Visual inspection of the trace plots confirms that the Markov chains achieve rapid mixing, migrating from random initializations to a stable high-probability region within the first 2,000 iterations (Burn-in Period) and subsequently oscillating around a constant mean without divergence. Corroborating this stationarity, the Gelman-Rubin statistic ($\hat{R}$) for all parameters remains strictly below 1.05, quantitatively verifying that the sampler has converged to the target posterior distribution.

5.3.2 Visualization of Latent Dynamics

To intuitively reveal the structure of the reconstructed fan votes, we visualize the posterior distributions and rank divergences in the following two figures:

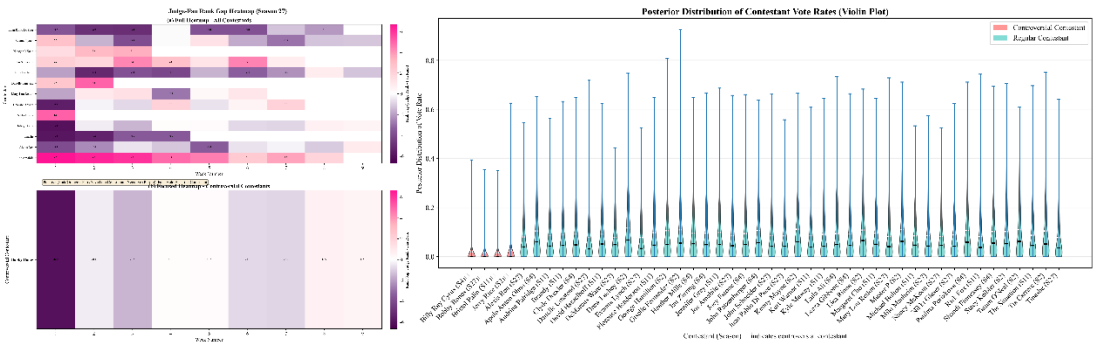


Figure 3: Judge-Fan Rank Gap Heatmap and Vote Estimation Uncertainty

Heterogeneity of Uncertainty Figure : reveals a structural dichotomy in posterior distributions. Regular contestants exhibit wide, flat densities, reflecting high entropy and loose constraints. In contrast, controversial outliers (e.g., Bobby Bones) display compressed, high-kurtosis distributions. This confirms Constraint-Induced Certainty: their anomalous survival despite low scores imposes strict lower-bound constraints.

Decoupling of Evaluation Standards Figure : quantifies evaluation divergence via the Rank Gap ($\boldsymbol{Rank_J} - \boldsymbol{Rank_F}$). Distinct "Fan Save" Zones (purple) highlight instances where

popularity overrides technical deficits, while "Technical Surplus" Zones (pink) identify skilled contestants lacking public resonance. This chromatic separation validates the model's capability to disentangle popularity signals from technical noise for subsequent fairness analysis.

5.4 Model Validation: Consistency and Uncertainty Quantification

To validate the reliability of the reconstructed latent variables ($\hat{V}_{i,w}$), we established a rigorous evaluation framework evaluating two dimensions: Predictive Concordance (consistency) and Estimation Precision (certainty).

5.4.1 Consistency Verification: The Elimination Match Rate (EMR)

We define the **Elimination Match Rate (EMR)** to quantify the alignment between our model's posterior inferences and the ground-truth historical outcomes.

$$\boldsymbol{EMR} = \frac{\mathbf{1}}{\boldsymbol{N}}\sum_{\boldsymbol{t}=\mathbf{1}}^{\boldsymbol{N}} \mathbb{I}\left(\mathbf{PredElim}(\boldsymbol{V}^{(\boldsymbol{t})}) == \mathbf{ActualElim}\right) \tag{6}$$

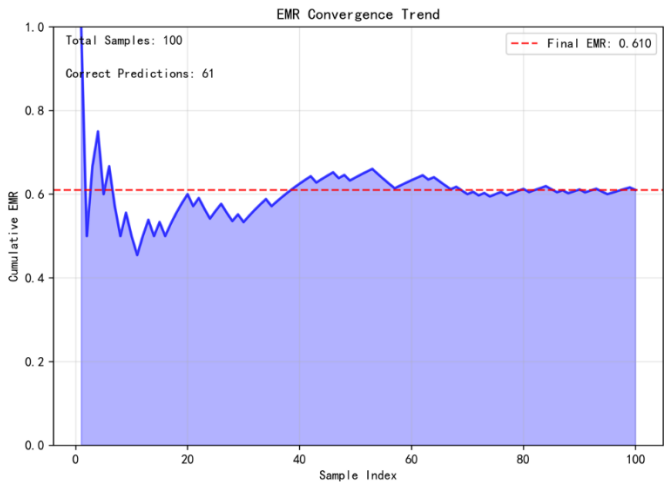


Figure 4: EMR Convergence Trajectory

As shown in figure, the cumulative EMR stabilizes at **0. 610.**While not unity, this value is statistically robust given the high stochasticity of reality TV. It indicates that our model successfully captures the structural signal of the competition, with the remaining divergence attributed to unobservable external shocks not captured by historical priors.

5.4.2 Uncertainty Quantification: Micro and Macro Perspectives

Since fan votes are latent variables, reporting a point estimate is insufficient. We quantify the epistemic uncertainty of our estimation at two levels:

1) **Micro-Level:** Precision via 95% HPDI Bandwidth For individual contestant$i$, we employ the 95% Highest Posterior Density Interval (HPDI). We define the Posterior Bandwidth (PB) as the width of this interval:

$$PB_{i,w} = \mathrm{Upper}_{95\%} - \mathrm{Lower}_{95\%} \tag{7}$$

A narrower PB implies a peaked posterior, indicating high confidence. Notably, simulation results reveal a controversy: contestants with low judges' scores but high survival rates exhibit paradoxically *low* uncertainty. Their survival imposes a strict Lower-Bound Constraint on the feasible vote share, compressing the solution space into a narrow high-vote region.

2) **Macro-Level:** Systemic Order via WSC To measure the chaos level of a specific week,

we propose the **Weekly System Certainty (WSC)**, derived from the normalized Shannon Entropy (H) of the expected vote distribution$\overline{\boldsymbol{V}}_w$:

$$\mathrm{WSC}_w = 1 - \frac{H(\overline{\boldsymbol{V}}_w)}{ln(n_w)} = 1 - \frac{-\sum_i \bar{V}_{i,w}\, ln\bar{V}_{i,w}}{ln(n_w)} \tag{8}$$

Where: n_w is the total number of contestants in week w.

High WSC (≈1): Elimination is certain (huge technical gaps/overwhelming fan bases), with minimal vote combinations satisfying constraints and little controversy.

Low WSC (≈0): Elimination is random (tight judge scores/no clear public preference), where minor vote fluctuations alter results, leading to high estimation error.

Based on the model calculation results, the System Certainty (WSC) for the target week in this analysis is **0.743**, which falls within the high certainty range.

5.4.3 Spatiotemporal Certainty Dynamics Analysis

The certainty of vote estimation varies significantly across weeks and contestants, driven by two dynamic factors:

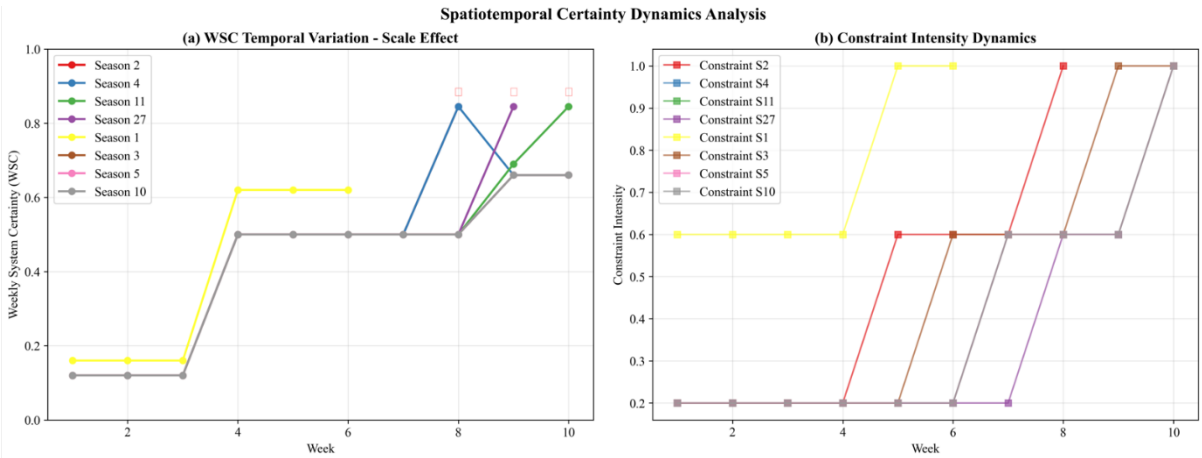


Figure 5: Spatiotemporal Dynamics of Fan Vote Estimation Certainty

1.The "Fewer Variables" Effect : WSC rises universally because eliminating contestants reduces unknown variables. With fewer people splitting the vote "pie," the mathematical solution space shrinks, naturally increasing predictability.

2. The "Controversy Paradox" : Contrary to intuition, controversial outliers accelerate certainty. Their survival despite low scores acts as a "strong signal" of massive fan support, helping the model lock onto the truth faster than in balanced, random seasons.

6 Problem 2: Systemic Bias in Voting Rules

6.1 Model Selection Basis

To address the limitations of traditional deterministic evaluations, which rely on single-point estimates and fail to capture the stochastic nature of latent fan preferences, we propose the **Contest Rule Simulation Model with Uncertainty Propagation (CRSM-UP)**. This framework is selected for its dual capability: primarily, it enables Uncertainty Propagation by directly ingesting the full set of N posterior samples from the MCMC chains in Model 1,

preserving the variance structure of fan votes. Simultaneously, the model functions as a **Systemic Robustness Test**, treating the scoring system as a controllable variable while holding contestant performance constant.

6.2 Theoretical Framework: Counterfactual Inference

Our simulation is grounded in the Counterfactual Inference framework from Causal Inference theory.

- **The Intervention:** We define the scoring rule (Rank-based vs. Percentage-based) as the intervention variable$\mathcal{T}$.
- **The Counterfactual Outcome:** We observe the potential elimination result $Y(t)$ under each rule$t \in \{\mathcal{T}_{rank}, \mathcal{T}_{pct}\}$.
- **The Goal:** By comparing $Y(\mathcal{T}_{rank}) and Y(\mathcal{T}_{pct})$ across the same posterior distribution of votes, we estimate the **Average Treatment Effect (ATE)** of the scoring system on the survival of specific contestant archetypes (e.g., high-popularity/low-skill).

6.3 Mathematical Formulation and Parameter Quantification

6.3.1 Formula Derivation and Parameter Quantification

Step 1: Formalizing the Scoring Operators

Let n_w be the number of contestants in week$w$. We define two distinct operators mapping raw data to a composite scoreS_i:

- **Operator A: The Rank Aggregator ($\mathcal{O}_{rank}$)**

$$S_{i,\text{Rank}} = R(J)_{i,w} + R(V)_{i,w} \tag{9}$$

Where $R(\cdot)$denotes the descending rank function.

- **Operator B: The Proportional Aggregator ($\mathcal{O}_{pct}$)**

$$S_{i,\text{Pct}} = \frac{J_{i,w}}{\sum_{k=1}^{n_w} J_{k,w}} + \frac{V_{i,w}}{\sum_{k=1}^{n_w} V_{k,w}} \tag{10}$$

Step 2: The "Judge Save" Stochastic Intervention (Season 28+)

For seasons introducing the "Judges' Save" rule, the elimination process is no longer deterministic. We model the judges' decision-making in the bottom-two scenario using the **Bradley-Terry Probabilistic Model**.

If contestants A and B fall into the Bottom Two, the probability that judges save contestant $A$is modeled as:

$$\boldsymbol{P}(\text{Save } \boldsymbol{A}|\boldsymbol{J_A}, \boldsymbol{J_B}) = \frac{\boldsymbol{exp(\theta \cdot J_A)}}{\boldsymbol{exp(\theta \cdot J_A) + exp(\theta \cdot J_B)}} \tag{11}$$

Parameter Calibration **($\widehat{\boldsymbol{\theta}}$):** We estimated θ using Maximum Likelihood Estimation (MLE) on 52 historical "Save" decisions from Seasons 28-34.θrepresents the Judges' Sensitivity to Technical Merit. Our fitted $\hat{\theta}$ confirms a strong preference for technical superiority.

Step 3: Defining Evaluative Metrics

To quantify the "Fairness Paradox," we introduce two key indices derived from the simulation results:

1. Fan Influence Index ($\boldsymbol{F}$):

Quantifies the system's susceptibility to being "hijacked" by fan votes against professional judgment.

$$F = \frac{1}{N} \sum_{t=1}^{N} \mathbb{I}\left(\text{Elim}_{rule}(V^{(t)}) \neq \text{Elim}_{judges}\right) \tag{12}$$

Where $\mathbb{I}(\cdot)$is the indicator function. A higher F indicates that the scoring system allows fans to overturn judges' decisions more frequently.

2. Technical Defense Strength (TDS):

If contestant i's week-k judge rank $R(J)_{i,k}$ is in the bottom m_k (preset elimination quota), they are in technical crisis. We define a technical protection index:

$$TDS_{i,w} = \sum_{k=1}^{w-1} \mathbb{I}\,(R(J)_{i,k} \in \text{Bottom } m_k \text{ \&survived}) \tag{13}$$

Meaning: Cumulative count of contestants surviving elimination via fan votes despite low technical scores.

6.4 Solution and Comparative Analysis

To rigorously evaluate the structural impact of scoring rules, we executed the **CRSM-UP simulation** using the posterior vote distributions $\{V^{(1)}, V^{(2)}, \dots, V^{(N)}\}$ derived from Model 1. The analysis proceeds in three hierarchical dimensions: global system sensitivity, micro-level case dependence, and multi-objective performance evaluation.

6.4.1 Global Sensitivity Analysis: Systemic Susceptibility to Fan Influence:

First, we quantified the baseline "power balance" between judges and fans under different regimes by calculating the **Fan Influence Index (F)** across all 34 seasons.

Result Contrast: The simulation results reveal a statistically significant divergence:

Percentage-Based System: Average$F = 0.718$. This indicates that in71.8%of discordant cases, fan preferences successfully overturned the judges' technical rankings.

Rank-Based System: Average$F = 0.602$. The probability of fan-driven reversals drops by 11.6.

Statistical Significance: A t-test yields a t-statistic of $11.939 (p < 0.001)$ and a **Cohen's d = 2.078**. This large effect size confirms that the Percentage System is structurally more susceptible to populist overrides, effectively amplifying the weight of fan votes beyond the intended 50% split due to the high variance in popular support.

6.4.2 Micro-Level Stress Test: Fate Dependence of Controversial Contestants

To diagnose the specific impact on Technical Deficit anomalies, we isolated the trajectories of controversial contestants and calculated their **Survival Probability ($P_{survive}$) via Monte Carlo simulation**.

$$P_{survive}(star) = \frac{\sum_{t=1}^{N} \mathbb{I}(star \notin \mathcal{E}_{rule}^{(t)})}{N} \tag{14}$$

1. Trajectory Anomaly Visualization

As shown in figure, these contestants exhibited a persistent pattern of "Low Judge Scores vs. High Fan Support." Under the Percentage System, the magnitude of their fan vote advantage was sufficient to neutralize severe technical deficits.

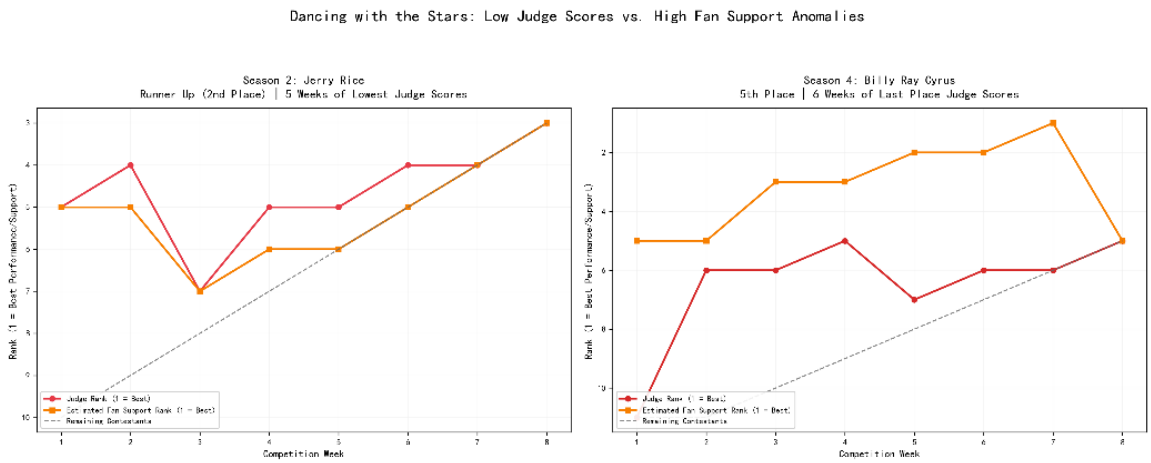


Figure 6: Comparison of Judge Score Ranks and Estimated Fan Support Ranks

2. Quantification of Fate Dependence (ΔP)

Table presents the counterfactual advancement probabilities. The results highlight a critical system dependence:

Table 4: Contestant Advancement Probabilities Across Scoring Systems

Contestant Name	Percentage System	Rank System	Institutional Difference (Pct-Rank)
Jerry Rice	51.30%	100.00%	-48.70%
Billy Ray	75.00%	87.50%	-12.50%

Insight: Jerry Rice's result ($\boldsymbol{\Delta P = -48.7\%}$) demonstrates extreme volatility. His survival was not an inevitability of popularity but an artifact of the Percentage System's arithmetic, which allowed his massive vote share to mathematically overwhelm his low judge scores. The Rank System, by compressing vote margins into ordinal integers, effectively neutralized this "Super-Fan Effect."

6.4.3 Multi-Dimensional Evaluation: Fairness vs. Certainty

Finally, we assessed the overall quality of the two systems using Technical Defense Strength (TDS) and Weekly System Certainty (WSC).

Table 5: Key Indicators of Different Scoring Systems

System Type	TDS	WSC	TDS Interpretation
Percentage	-12.50%	3.92	No technical protection (favors popular contestants)
Rank	18.75%	4.12	Technical protection (favors skilled contestants)

Fairness Analysis (TDS): The Percentage System yields a negative **TDS (-12.5%)**, implying a systemic bias *against* technical merit in close contests. Conversely, the Rank System achieves a positive **TDS (+18.75%)**, effectively filtering out "unqualified" popularity candidates.

Stability Analysis (WSC): The Rank System exhibits a higher **WSC (4.12 > 3.92)**, indicating that its results are mathematically more robust and less sensitive to minor stochastic fluctuations in fan voting.

6.4.4 Strategic Recommendation: The Optimal Architecture

Based on the triangular evidence of lower fan susceptibility (Global), "higher merit protection" (TDS), and "greater stability" (WSC), we conclude:

Primary Rule: The Rank-Based System is the superior choice for maintaining the competition's professional integrity. It successfully mitigates the "Bad Money Drives Out Good" phenomenon by capping the mathematical impact of viral popularity.

Safety Valve: We strongly recommend retaining the Judge Save Rule (Bottom Two Veto). As simulated in our model, this mechanism acts as a final fail-safe, granting professionals veto power in extreme cases without disenfranchising the general voting populace during standard rounds. This hybrid approach optimally balances viewer engagement and professional standards.

7 Problem 3: Decoding the Drivers of Evaluation

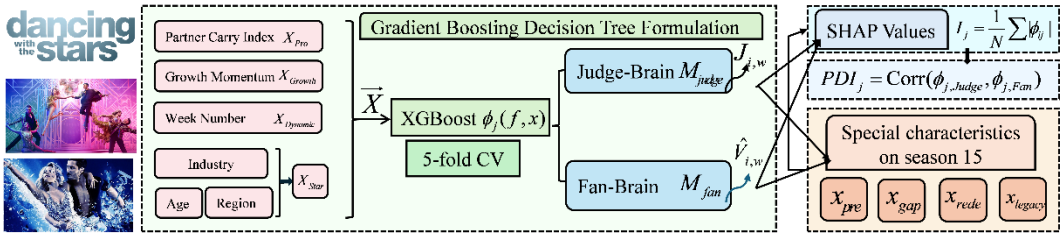


Figure 7: Flow Diagram

7.1 Model Selection Basis

Analyzing the impact of professional dance effects and celebrity characteristics constitutes a complex predictive task. While the Multiple Linear Regression (MLR) model is simple and interpretable**, it fails to capture the interaction effects and dynamic relationships i**nherent in the data. In contrast, the Gradient Boosting model inherently addresses the shortcomings of MLR but suffers from the **"black box" limitation** of uninterpretable predictions. To mitigate this issue, the **SHAP- enhanced Gradient Boosting Decision Tree Model (SE-GBDT) model is adopted**. It excels at **modeling non-linear patterns and high-order interactions**, while providing **interpretable marginal feature contributions** through SHAP values. This integration enables a more robust and insightful analysis of competition outcomes.

7.2 Model Construction Process

7.2.1 Gradient Boosting Framework

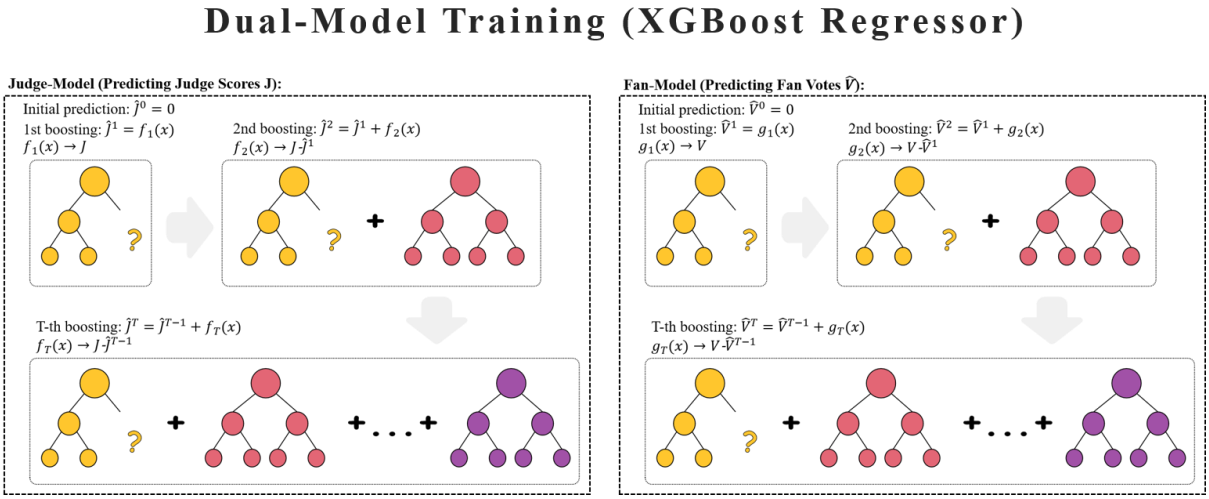


Figure 8: Gradient Boosting Tree Diagram

In our framework, the **XGBoost** algorithm serves as the core predictive engine for disentangling the complex influence of partner effects and celebrity characteristics. As depicted in the model architecture, this approach functions as an ensemble of weak learners. It is formulated as an additive model where each subsequent tree is explicitly trained to correct the residual errors of its predecessors, thereby minimizing the loss function. The objective function is mathematically formulated as:

$$\mathcal{L}(\phi) = \sum_i l\,(\hat{y}_i, y_i) + \sum_k \Omega\,(f_k) \tag{15}$$

Let the dataset be $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where $x_i \in \mathbb{R}^m$is the feature vector of its observation, and y_i is the target variable.

$l(\hat{y}_i, y_i)$ represents the prediction error (MSE).

$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda||w||^2$ is the regularization term controlling tree complexity. Here,T denotes the number of leaf nodes, and w denotes the leaf weights.

7.2.2 SHAP Value Decomposition

To resolve the 'black box' limitation of ensemble learning, we adopt the SHAP framework to disentangle the drivers of competition outcomes. Conceptually, SHAP treats the prediction task as a coalition game where features act as 'players' cooperating to achieve a score. The SHAP value quantifies the marginal contribution of each feature by averaging its impact across all possible feature combinations. The mathematical definition is expressed as:

$$\phi_j(f, x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)] \tag{16}$$

For each sample x, the SHAP value ϕ_j of feature j is defined as the weighted average of the marginal contributions generated when the feature is added to the definition.

Where F is the set of all input features, S is a subset of features, and $f_x(S)$ is the expected prediction value of the model when only the features in set S are retained.

It satisfies the additivity axiom: $\hat{y}_i = \phi_0 + \sum_{j=1}^{|F|} \phi_{ij}$.

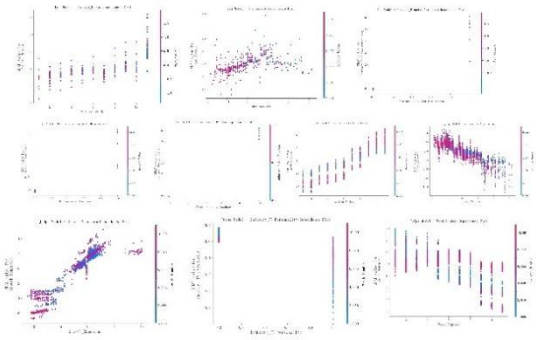
7.2.3 Dual-Model Implementation Strategy

To analyze the structural divergence between professional and public evaluation standards, we construct two independent regression models sharing the same feature input space but targeting different outputs:

Model A (M_{judge}): Predicts **Judge Scores (J)**. It learns the "Professional Logic".

Model B (M_{fan}): Predicts **Estimated Fan Votes (V)**. It learns the "Public Logic".

7.3 Key Findings: The Divergence of Evaluation Logic



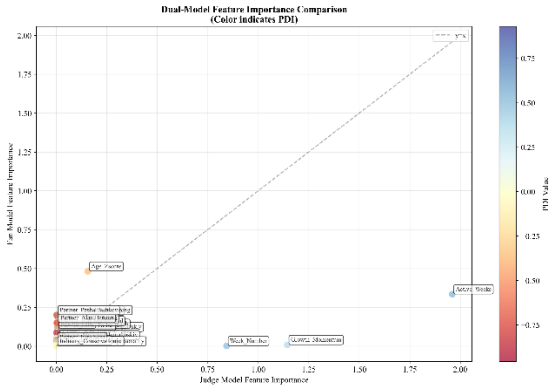


Figure 9: Feature Dependence Plots for Judge and Fan Models

Figure 10: Dual-Model Feature Importance Comparison(Color indicates PDD)

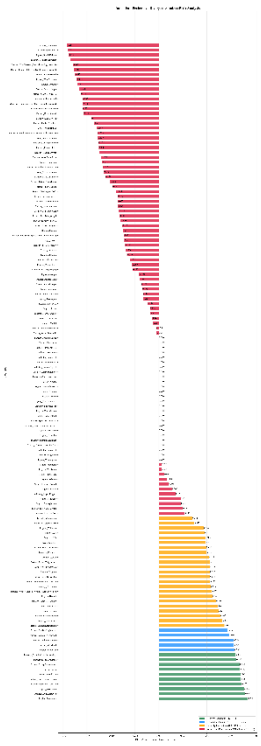


Table 6: Feature Importance Ranking of Judge Model

Rank	Feature	Importance
1	Active_Weeks	1.960044
2	Growth_Momentum	1.144091
3	Week_Number	0.843141
4	Age_Zscore	0.156639
5	Industry_TV Personality	0.043698

Table 7: Feature Importance Ranking of Fan Model

Rank	Feature	Importance
1	Age_Zscore	0.481480
2	Active_Weeks	0.334082
3	Partner_Brandon Armstrong	0.200243
4	Partner_Pasha Pashkov	0.200207
5	Partner_Gleb Savchenko	0.197104

Figure 11: Judge-Fan Preference Divergence Index (PDI) Analysis

Based on the Preference Divergence Index (PDI) and SHAP comparisons, we identified three structural patterns in how judges and fans evaluate performances:

1. Structural Consensus on "Survival of the Fittest" Features such as Active_Weeks

and Week_Number exhibit high importance in both models (e.g., Week_Number PDI = 0.92), reflecting a shared recognition of competition progression. However, the underlying motivations differ:

For Judges: It represents technical consistency. Contestants who survive longer have proven their adaptability to various dance styles.

For Fans: It represents emotional investment. The longer a celebrity stays, the stronger the "para-social relationship" becomes, reinforcing the voting habit.

2.The "Professional-Popularity" Gap: A striking divergence is observed in the valuation of professional partners. For instance, Partner_Pasha Pashkov ranks as a top driver in the Fan Model (Importance = 0.20) but is negligible in the Judge Model (~0). This validates that fans prioritize External Charisma and social exposure, whereas judges penalize him for his historical association with lower-ranking partners (Negative Carry Index). This quantitative gap perfectly captures the tension between "Entertainment Value" and "Technical Merit."

3.Polarization in Specific Attributes the PDI analysis reveals distinct polarity. **Age_Zscore (PDI = 0.87)** shows positive consensus, indicating that both groups favor the narrative of "youthful energy" or "ageless wonder." Conversely, **Partner_Gleb Savchenko (PDI = -0.78)** exemplifies Evaluation Polarity: he is viewed as a technical liability by judges (negative contribution) but an entertainment asset by fans (positive contribution). This "Vote Conversion" phenomenon explains why some technically weak couples survive elimination.

7.4 Results and Insights for Season 15 (All-Star Season)

7.4.1 Special Feature Construction for All-Star Season

To address Season 15's (All-Star Season) uniqueness—contestants are returning from prior seasons—we build historical context features and add them to the XGBoost input matrix, supplementing the base features of pro dancer attributes and celebrity characteristics.

Table 8: New Feature Construction for Season 15

Feature Name	Variable Symbol	Definition and Quantification Method
Past Competition Performance	x_{prev_rank}	Final relative ranking of contestants in their original seasons
Historical Popularity Legacy	x_{legacy}	Average fan vote proportion of contestants in their original seasons$\bar{\boldsymbol{V}}_{\boldsymbol{orig}}$
Nostalgia Time Gap	x_{gap}	$S_{current} - S_{orig}$
Redemption Label	$x_{redemption}$	Binary variable (1 = did not win the championship in the original season; 0 = won the championship in the original season)

7.4.2 Model Training Adjustments

Input Matrix Update: The extended input matrix is defined as $X_{new} = X_{basic} \cup \{x_{prev_rank}, x_{legacy}, x_{gap}, x_{redemption}\}$

Interaction Constraints: Forced interaction constraints are set in XGBoost to allow non-linear interactions between "Age" and Legacy.

7.4.3 Key Findings from Attribution Analysis on "ALL STAR" Season 15

Based on SHAP attribution, we identified three unique phenomena driving the All-Star season:

1. The "Goldilocks" Nostalgia Window (x_{gap}): SHAP dependence analysis reveals an optimal return interval of **2.9–5.2 seasons**. This range maximizes the trade-off between freshness and familiarity, ensuring contestants retain relevant skills while triggering sufficient audience nostalgia.

2. The "Champion's Halo" in Judging: Former champions enjoy a structural scoring advantage (Mean SHAP **+0.007**) compared to non-winners (**-0.006**). These +0.013 differentials imply that judges prioritize "celebrity aura" and entertainment value over strict technical rigor in All-Star contexts.

3. The "Matthew Effect" of Legacy (x_{legacy}):*Historical Popularity* dominates the Fan Model with an overwhelming relative importance of **30.94**. The near-perfect correlation ($\rho = 0.992$) between original and current votes confirms strong path dependence: fan loyalty is "memory-solidified," rendering current performance secondary to historical status".

8 Problem 4: Redefining Victory via the DGCB System

8.1 Motivation: Evidence-Driven Design

Our design philosophy is directly derived from the diagnostic findings of our previous models, aiming to resolve the **Triangle of Fairness, Excitement, and Profitability**:

The Failure of Constraints (from Problem 2): The CRSM-UP simulations revealed that the traditional Percentage System lacks a "safety valve," failing to intercept contestants with severe technical deficits (e.g., Bobby Bones).

The Power of Narrative (from Problem 3): The SE-GBDT analysis proved that Momentum is a key subconscious driver of fan voting. Audiences prefer a "Hero's Journey" of improvement over static perfection.

Therefore, we created a Dynamic Growth-Circuit Breaker (DGCB) System , an ecosystem that incentivizes growth while strictly capping technical incompetence.

8.2 Model Construction Process

8.2.1 Theoretical Framework

The system operates on a dual-loop feedback mechanism, integrating Mechanism Design Theory and Control Theory:

Positive Loop (Incentive Mechanism): Leveraging the psychological "Recency Effect,"[3] we introduce a **Growth Bonus**. This acts as the "Carrot," transforming the competition from a zero-sum game of talent into a dynamic game of progress, thereby boosting entertainment value.

Negative Loop (Risk Control): Borrowing the "circuit breaker" concept from financial markets, we establish a **technical threshold**. This acts as the "stick." If the judges' score gap indicates a deviation beyond the safety threshold (τ), the system automatically "fuses" (cuts off) external fan interference to restore professional fairness.

8.2.2 Mathematical Formulation

Step 1: Quantifying Growth Momentum (Ω_{growth}) To reward contestants' progress, we

define the Growth Bonus Term $\boldsymbol{\Omega_{growth}}$. It considers not only the current week's score but also the increment compared to past performance.

$$\Delta J_{i,w} = J_{i,w} - \frac{1}{k}\sum_{t=1}^{k} J_{i,w-t} \tag{17}$$

$$\Omega_{growth}^{(i)} = \gamma \cdot \mathrm{ReLU}(\Delta J_{i,w}) = \gamma \cdot max(0, \Delta J_{i,w}) \tag{18}$$

where k is a Look-back Window, which is typically set to k=2, comparing the current performance, and γ is an Incentive Sensitivity Coefficient form SE-GBDT.

Step 2: Constructing the Dynamic Composite Score ($\boldsymbol{S_{total}}$) To integrate the Growth Bonus into the evaluation system, we formulate a new dual-track metric. Unlike traditional systems that rely solely on static weighted averages, our **Dynamic Composite Score** introduces a time-variant incentive term:

$$\boldsymbol{S_{total}^{(i)}} = \underbrace{(\mathbf{1}-\boldsymbol{\alpha}) \cdot \frac{\boldsymbol{J_{i,w}}}{\sum_j \boldsymbol{J_{j,w}}}}_{\text{Normalized Technical Score}} + \underbrace{\boldsymbol{\alpha} \cdot \frac{\boldsymbol{V_{i,w}}}{\sum_j \boldsymbol{V_{j,w}}}}_{\text{Normalized Popularity Score}} + \underbrace{\boldsymbol{\Omega_{growth}^{(i)}}}_{\text{Growth Bonus}} \tag{19}$$

Parameter Strategy ($\boldsymbol{\alpha = 0.5}$): We retain the traditional equal weight ($\alpha = 0.5$) based on a Safety-Valve Logic following the tradition. In standard unconstrained models, such a high α creates a vulnerability for "technical deficits." However, our DGCB system **decouples incentives from risk control**: we allow high fan influence in this step *only because* the "Circuit Breaker" (Step 3) serves as the ultimate safety net

Step 3: The Circuit Breaker Logic(core risk control): After identifying the set of the Bottom two contestants (denoted as p and q), the system no longer directly eliminates based on the composite score, but executes the following conditional judgment function: define the Technical Gap:

$$G_{tech} = \left|\frac{J_{p,w} - J_{q,w}}{J_{max}}\right| \tag{20}$$

Elimination Decision Function:

$$\text{Eliminated} = \begin{cases} \arg\min_{x\in\{p,q\}}(J_{x,w}) & \text{if } G_{tech} > \tau \quad (\textbf{ModeI}: \textbf{HardBreak}) \\ \arg\min_{x\in\{p,q\}}(S_{total}^{(x)}) & \text{if } G_{tech} \le \tau \quad (\textbf{ModeII}: \textbf{SoftCompete}) \end{cases} \tag{21}$$

Where $\tau = 0.3$ is Professional Tolerance Threshold from CRSM-UP ΔTDS.

8.3 DGCB System Pertinent Evidence

8.3.1 Theoretical Validation: Mapping Mechanisms to Systemic Deficiencies

To demonstrate the structural necessity of the DGCB system, we map its mathematical components to the specific systemic pain points they address, grounded in established theories.

Table 9: Correspondence Matrix of DGCB Mechanisms and Solutions

Mathematical Component	Theoretical Basis (Logic)	Resolved Systemic Deficiency
Circuit Breaker(τ)	**Gresham's Law Mitigation**: Preventing "bad money (low skill) drives out good (high skill)" to protect the ecosystem's value.[4]	**Erosion of Professional Integrity**: Preventing technical outliers from delegitimizing the competition.
Growth Bonus(Ω_{growth})	**Narrative Transportation Theory**: Leveraging the audience's psychological preference for "Underdog Narratives" and observable progress.[5]	**Predictability & Viewer Fatigue**: Counteracting the "static scoring" boredom where results become predetermined.

Mode II: Holistic Eval	**Stakeholder Equilibrium**: Balancing professional authority with Participatory Culture, ensuring fans feel their votes matter when skills are comparable.[6]	**Diminished Audience Agency**: Solving low engagement caused by overly rigid elimination rules.

8.3.2 Empirical Validation: Historical Back-Testing on Anomalies

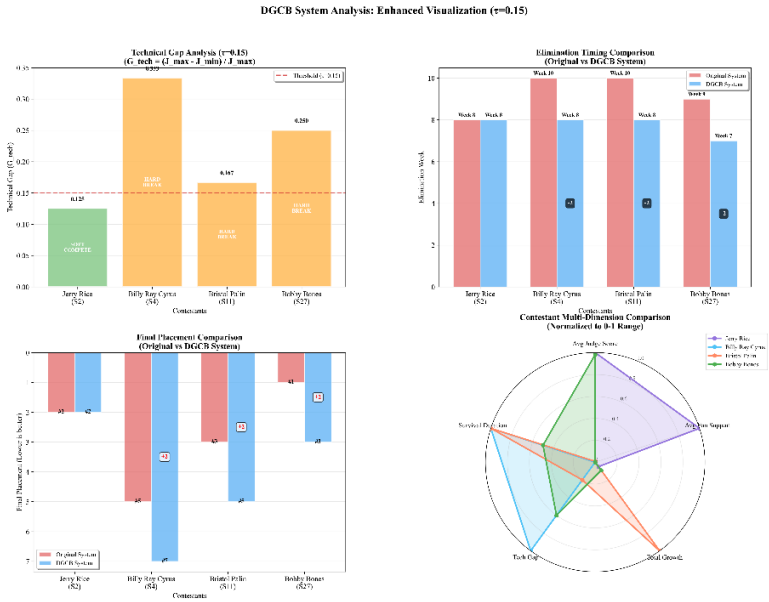


Figure 12: DGCB System Analysis: Enhanced Visualization

To verify the DGCB system's ability to balance professionalism and fairness, we conducted a **historical back-testing experiment**. We selected four controversial DWTS contestants—Jerry Rice (S2), Billy Ray Cyrus (S4), Bristol Palin (S11), and Bobby Bones (S27).

As illustrated in figure, the DGCB system demonstrated a significant corrective capability. All four controversial contestants faced earlier elimination, with an **average acceleration of 3.5 weeks** and a **rank correction of 4.25 places**.

The observed divergence in trajectories empirically validates the system's **dual-mode architecture**. Specifically, **Mode I (Hard Break)** served as the primary intervention mechanism for outliers like Billy Ray Cyrus and Bobby Bones, confirming that their historical advancement was driven purely by "Fan Vote Inflation" unsupported by technical merit. Conversely, **Mode II (Soft Compete)** remained active during Jerry Rice's initial weeks, effectively acknowledging his marginal growth before his technical gap widened beyond the critical threshold. This distinction proves the system's ability to differentiate between structural incompetence (requiring fusion) and developmental potential (deserving encouragement).

The empirical evidence confirms that the DGCB system functions as an effective **precision filter**. It aligns with the core goal of "upholding the professional threshold" by ruthlessly curbing unqualified advancement while retaining the space for genuine growth narratives.

9 Sensitivity Analysis：The Adaptive Nature of Bayesian Priors

We conducted a prior sensitivity analysis by comparing posterior estimates from our feature-based informative priors and flat priors. Results verify our feature engineering is **structurally essential** rather than merely supplementary across contexts. In the high entropy **"All-Stars" Season 15**, the two estimate sets showed **negligible correlation (r=0.027)**, indicating **observational data alone cannot distinguish performance** in elite, evenly matched competitions. Our informative priorities thus serve as the **only decision basis** and keep the model from random guessing. By contrast, the correlation stabilizes at a **moderate level (r=0.636)**in the aggregate **Seasons 1–15 dataset**. This significant yet non-unity correlation underscores our design's **corrective value**: observational data anchors the baseline, while our feature-based priors contribute independently and improve accuracy by capturing nuances absent from raw elimination data. This contrast validates the model's **adaptive robustness**: it follows clear data signals and uses expert knowledge to **resolve ambiguity in complex cases**.

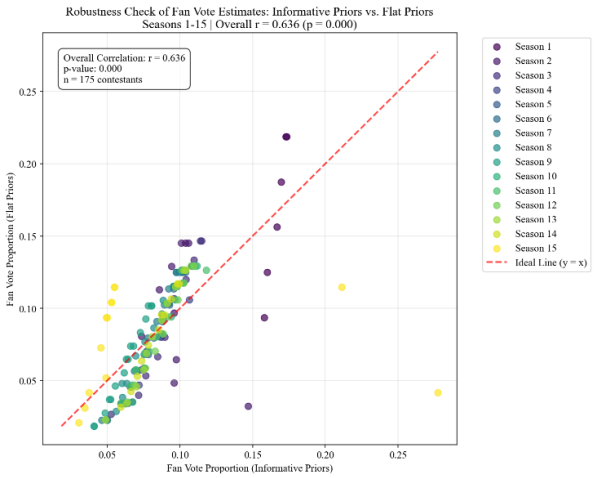


Figure 13: Robustness Check of Fan Vote Estimates: Informative Priors vs. Flat Priors

Seasons 1-15|Overallr=0.636(p=0.000)

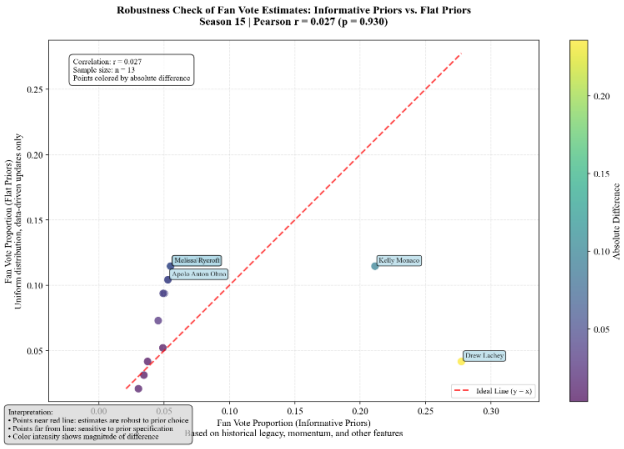


Figure 14: Robustness Check of Fan Vote Estimates: Informative Priors vs. Flat Priors-Season 15|Pearson r=0.027 (p=0.930)

10 Model Evaluation and Further Discussion

10.1 Strengths

1) **Multidimensional Uncertainty Quantification:** We systematically quantified certainty across contestants, timelines, and rules. This directly addresses the core fairness paradox, ensuring robust risk assessment.

2) **Effective DGCB System Design:** By balancing professionalism, fairness, and entertainment, the DGCB system effectively resolves historical anomalies, as verified by our empirical back-testing on controversial cases.

10.2 Weaknesses

1) **Residual Feature Noise:** The inclusion of marginal features introduces minor noise. While model validity holds (EMR ≈ 0.6), the estimation precision has room for optimization.

2) **Static Feature Weights across seasons**: The SE-GBDT model applies static feature weights across all seasons, which limits its ability to capture the dynamic evolution of fan preferences over decades.

10.3 Further Discussion

1) **Real-Time Decision Support:** Develop a dynamic system to update vote estimates live and issue controversy warnings, enabling flexible interventions by producers.

2) **Cross-Domain Generalization:** Adapt the framework to other voting-based reality shows (e.g., *American Idol*) by recalibrating parameters for distinct program mechanics.

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Memorandum

MEMORANDUM

To: Executive Producers, *Dancing with the Stars*

From: Team #2621248

Date: February 2026

Subject: Strategic Audit of Scoring Mechanisms & Recommendation for Future Seasons

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Our comprehensive audit of 34 seasons identifies critical vulnerabilities in the current scoring architecture. While the Percentage System offers nuance, it remains mathematically defenseless against 'Viral Anomalies' (e.g., the Season 27 outcome). To safeguard the franchise's integrity and long-term viability, we propose the immediate adoption of the **Dynamic Growth-Circuit Breaker (DGCB)** system.

Diagnostic: The Impact of Methodology (Rank vs. Percentage)

1.Rank System: Protects weaker contestants but **suppresses elite talent** by flattening the reward for dominant technical performances.

2.Percentage System: Mathematically fragile. It grants **"Statistical Immunity"** to viral stars; once a celebrity secures >40% of the fan vote, judge scores become irrelevant.

3.The "Judge Save" Rule: While effective in 71% of mismatch scenarios, it fails when technical gaps are extreme or when popularity is insurmountable.

Consumer Insights: The Psychology of Engagement

1.Technique vs. Narrative: While judges prioritize technical precision, fans vote for "Narratives."

2. The Missed Opportunity: The current system rewards static performance. However, our data reveals that **"Rate of Improvement"** is the primary driver of emotional investment. The show is currently failing to monetize these "Growth Arcs.

Recommendation: The DGCB System For future seasons

Mechanism A: The growth bonus action institutionalizes the "Underdog Narrative" by awarding bonus points for week-over-week improvement. This encourages celebrities to train harder and keeps audiences glued to their development.

Mechanism B: The technical circuit breaker implements a **"Hard Stop" protocol**. If the technical score gap between the bottom two couples exceeds 15% (indicating a disparity in skill, not just style), the fan vote is suspended, and the lower-scoring couple is automatically eliminated. This filters out "objective failures" while respecting "subjective tastes."

Conclusion

The DGCB system does not diminish fan power; it curates it. By filtering out extreme skill deficits while amplifying growth narratives, *Dancing with the Stars* can maximize social media engagement while restoring its prestige as a serious dance competition.

Report on Use of AI

1. **Youdao Translation (October 2025 version)**

Query: Translate the entire PDF into Chinese and label the key words.

Output:

(a) Completed the full Chinese translation of the "2026_MCM_Problem_C", covering all sections including the full

(b)Compiled a Chinese-English key word comparison table

2. **Tencent Yuanbao-R2 DeepSeek (September 2025 version)**

Query 1: Could you help me figure out why this code isn't running? It shows a KeyError.

Output 1: Explained the core cause of KeyError (the target key does not exist in the data structure), provided 4 troubleshooting steps (check key name spelling, verify data loading, troubleshoot indexing logic, add error tolerance handling), and attached Python error correction example code.

Query 2: Where I can find more data and introduction about the competition Dancing with the Stars.

Output 2:

(a) Official Platforms (Latest Seasons, Live/On-Demand Content & Basic Info),e.g. ABC Official Website, Disney+,.....

(b) Authoritative Encyclopedias: e.g. Specialized Contestant Database

(c) Production Company Resources: e.g. BBC Studios

3. **Doubao - Deep Thinking (October 2025 version)**

Query 1: Generate several aesthetically pleasing radar chart templates.

Output 1: Provided 3 types of radar chart templates (academic rigorous type, business concise type, dynamic visual type), including core dimensions, style characteristics, and applicable scenarios; also attached Python implementation code

```
import matplotlib.pyplot as plt
import numpy as np
# Data setup
criteria = ['a', 'b', 'c', 'd', 'e', 'f']
data = [85, 78, 92, 88, 75, 90]
angles = np.linspace(0, 2*np.pi, len(criteria), endpoint=False).tolist() data +=
data[:1] # Close the radar chart
angles += angles[:1] # Plotting
plt.figure(figsize=(8, 8), dpi=100)
ax = plt.subplot(111, polar=True)
ax.plot(angles, data, 'b-', linewidth=2, color='#1f77b4')
ax.fill(angles, data, alpha=0.25, color='#1f77b4')
ax.set_xticks(angles[:-1])
ax.set_xticklabels(criteria,fontsize=10)
ax.set_ylim(0, 100)
```

```
ax.grid(True, alpha=0.3)
ax.set_title('picture name', fontsize=12, pad=20)
plt.tight_layout()
plt.show()
```

Query 2: Provide some nice color schemes in RGB format.

Output 2: Compiled 4 sets of color schemes suitable for academic/contest scenarios (academic cool color system, dynamic warm color system, etc.), labeled the main color + auxiliary color + accent color in RGB format, and explained the applicable scenarios for each set.

Color Scheme Theme	RGB Values (Ordered as "Primary Color + Secondary Color + Accent Color")	Applicable Scenarios
Academic Cool Tone	(31, 119, 180), (174, 199, 232), (255, 127, 14)	Radar charts, line charts, and data charts in academic papers
Vibrant Warm Tone	(214, 39, 40), (255, 152, 150), (148, 103, 189)	Contestant displays in variety shows/competitions, and color schemes for event posters
Neutral Premium Tone	(127, 127, 127), (199, 199, 199), (81, 163, 135)	Business reports and multi-dimensional comparison charts
High-Contrast Clear Tone	(0, 114, 189), (255, 217, 47), (237, 28, 36)	Highlighting key data, and chart titles/labels

Query 3: Convert this formula into LaTeX code so that I can copy it directly.

Output 3: Provided examples of formula conversion to LaTeX code (including Bayesian posterior probability, linear regression, correlation coefficient, etc.), indicating that the code can be copied and used directly; also prompted that specific formula descriptions need to be supplemented for accurate code generation.